

Cloud security

By Hafsa Bensaid



Cloud computing means storing data, running applications and accessing services through the internet instead of local computers. Because information is stored online, it becomes a target for attackers.

Cloud security includes all techniques and tools used to protect this data .

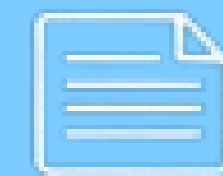
This research explains cloud security in detail: what it is, why it matters, its main threats, how to solve those threats, and real-world attack examples.



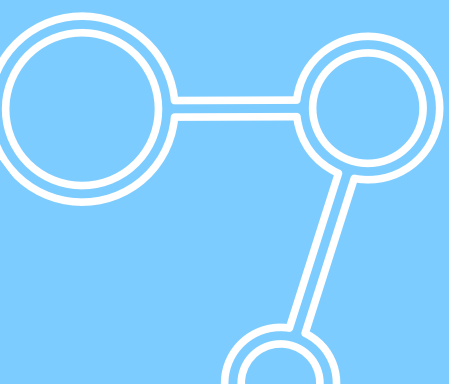
What is Cloud Security?

Cloud security refers to the combination of technologies, policies, controls, and practices used to safeguard cloud-based systems and data. It protects the confidentiality, integrity, and availability of information. Cloud security is based on the Shared Responsibility Model:

- Cloud providers (Google, Amazon, Microsoft) secure the physical infrastructure, networks, and hardware
- Customers secure their data, passwords, access rights, and settings.



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Understanding this model is essential because many cloud attacks happen due to user mistakes.



Importance of cloud security

Organizations, schools, banks, and governments now store huge amounts of data in the cloud. without cloud security :



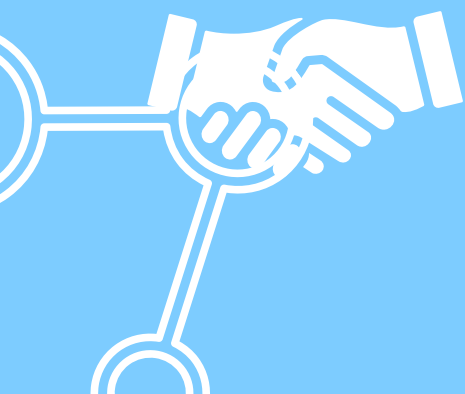
Hackers can steal personal details, financial information, or customer data.



Companies can lose availability of their services, affecting millions of users.



Attacks can cause financial loss, legal problems, and reputational damage.



Cloud security ensures stable service, trust between users and providers, and compliance with global regulations such as GDPR.





Cloud security threats

- **Data Breaches** :When attackers gain access to confidential cloud data. This can happen due to weak passwords, vulnerabilities,or bad settings.
- **Account Hijacking** : If a hacker steals login credentials, they can control cloud accounts and steal or delete data. Phishing attacks are a common cause.
- **Misconfiguration** :Wrongly configured cloud storage or security settings leave the system Exposed. This is one of the most common reasons for cloud attacks.
- **DdoS Attacks** :Hackers overload a cloud service with traffic, making it unavailable.
- **Insider Threats** :Employees or administrators intentionally or accidentally misuse their acces



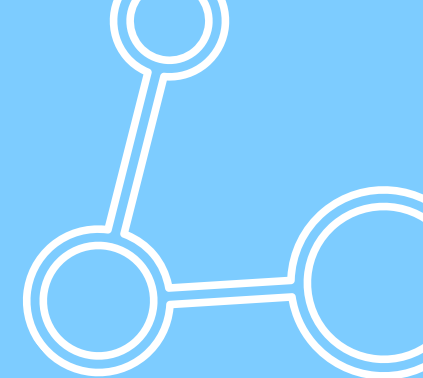
Solution to cloud threats

- **Multi-Factor Authentication (MFA):** Protects accounts even if passwords are stolen by requiring a second verification step.
- **Encryption:** Data is converted into unreadable code. Even if hackers access it, they cannot Read it.
- **Least Privilege Access:** Each user receives only the permissions they need reducing damage If an account is compromised
- **Regular Configuration Audits:** Ensures no storage bucket is left open and all settings followSecurity rules.
- **Continuous Monitoring:** Detects suspicious behaviour instantly, allowing quick response to Threats.
- **Firewalls & Intrusion Detection Systems:** Block malicious traffic and detect unusual activities.





Real world attacks

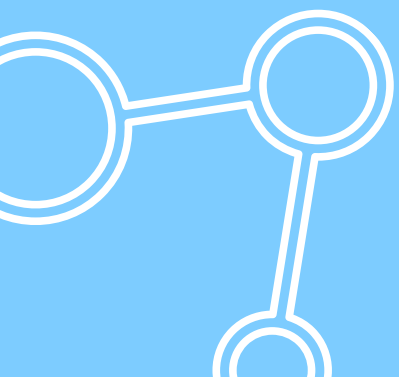


Capital One Data Breach (2019):

A misconfigured AWS firewall allowed an attacker to access highly sensitive customer data of more Than 100 million people.The attacker used a simple command to exploit the misconfiguration and download the data.

How it was solved:

Capital One fixed the misconfiguration, improved access rules, and invested in continuous Monitoring tools. AWS updated its security documentation to help prevent similar mistakes.

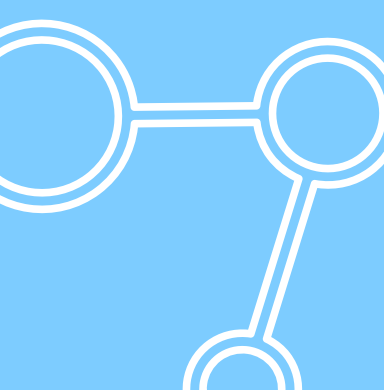


Dropbox Password Leak (2012):

Hackers stole the password of a Dropbox employee, gaining access to internal project files.

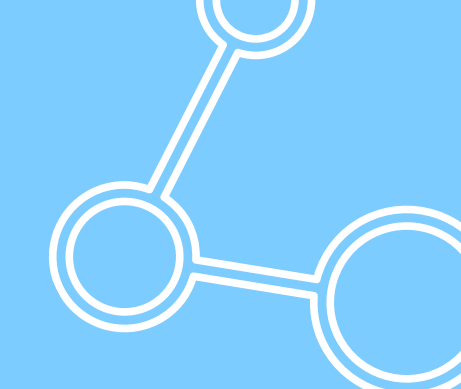
How it was solved:

Dropbox added multi-factor authentication, improved employee training, and enforced stronger password rules.



Conclusion

Cloud security is essential because modern life depends on cloud services for communication, Work, education, and storage. Although cloud environments face many threats, strong security techniques can greatly reduce risk. Understanding threats, applying solutions, using modern methods, and learning from real attacks Makes cloud environments safer for everyone.



Thank you

Any questions?

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